5	**(a)**	very high/infinite resistance for negative voltages up to about 0.4 V	B1	
		resistance decreases from 0.4 V	B1	[2]
	(b)	initial straight line from (0,0) into curve with decreasing gradient but not to horizontal	M1	
		repeated in negative quadrant	A1	[2]
	(c)	**(i)** $R = 12^2 / 36 = 4.0\,\Omega$	A1	
		or $I = P / V = 36 / 12 = 3.0$ A and $R = 12 / 3.0 = 4.0\,\Omega$	(A1)	[1]

(ii)	lost volts = $0.5 \times 2.8 = 1.4$ (V)	or $E = 12 = 2.8 \times (R + r)$	C1	
	$R = V/I = (12 - 1.4)/2.8$	or $(R + r) = 4.29\ \Omega$	C1	
	$= 3.8\ (3.79)\ \Omega$	or $R = 3.8\ \Omega$	A1	[3]

(d) resistance of the lamp increases with increase of V or I B1 [1]